

Rajendran Prathibha Devkar

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Software Engineer with 4+ years of experience building full-stack applications and ML systems, focused on making LLM applications measurable and trustworthy through confidence calibration, hallucination detection and agent observability (PyTorch, Ollama, FastAPI, React).

EXPERIENCE

Research Assistant, SEAR Lab - The University of Texas at Arlington

Apr 2026 - Present

- Consolidated a 6-service Docker stack (MQTT, InfluxDB, Grafana, a FastAPI data hub and Python collectors) into a single FastAPI app with an embedded dashboard, so the system now runs as one process instead of six services.
- Integrated live telemetry from three device APIs (LG ThinQ AC, Rheem EcoNet water heater, Enphase solar) into one REST API that computes a real-time Net Load metric, refreshed every 60 seconds by a background collector.

Software Developer Intern, Board Americas - Remote, Texas

May 2025 - Mar 2026

- Resolved 45+ medium-to-high severity bug tickets over 11 months on an enterprise forecasting platform (Aurelia/TypeScript, C#/.NET), tracing each issue end to end across UI, service calls, backend and configuration to ship minimal root-cause fixes.
- Fixed fiscal-year and date-range logic behind D3.js forecast charts (axis labels, period alignment, actual-vs-forecast rendering), and added validation and metadata checks so unsupported inputs show a clear error instead of breaking reports.
- Changed cache invalidation from per-user to per-company scope so forecast updates reach every user without a manual refresh, and fixed a null-date datepicker bug that was causing 500 errors in production.

Technology Analyst, Infosys (promoted twice from System Engineer) - Trivandrum, India

Oct 2019 - Nov 2022

- Developed React Native UI and integrated backend APIs for a curbside-pickup feature in the Tractor Supply Company mobile app, improving user satisfaction by 30% based on pilot-release feedback.
- Built RESTful APIs in Node.js for Infosys Meridian, a virtual events platform for meetups and webinars (200+ participants), and for an e-learning platform handling class scheduling, participation and registration across student, teacher and admin roles.
- Designed the e-learning platform's PostgreSQL schemas; when JMeter load tests (1K-10K virtual users) exposed server crashes in its calendar API under concurrent use, optimized the multi-table SQL joins for a team-reported 40% faster data retrieval.
- Established API testing standards and documentation for 20+ APIs.

PROJECTS

Confidence-Aware Meeting Intelligence - Python, Ollama, DistilBERT, Streamlit

- Built a meeting summarizer and action-item extractor that scores how far each extracted item can be trusted, pairing a local LLM (Ollama, llama3.2:3b) with an independently fine-tuned DistilBERT classifier.
- Calibrated the score via logistic regression over four signals (quote grounding, owner attribution, cross-model agreement, self-reported confidence) on 228 hand-labeled examples; the top tier nearly doubled held-out precision (58% vs. 35% unfiltered).
- Extracted the calibration logic into confidence-referee, a standalone MIT-licensed Python package (13 tests, CI on Python 3.9-3.12); on a second domain, RAG hallucination detection on HaluEval, one unmodified signal reached 0.97 AUC and 94% held-out accuracy.

Recurrent Depth Transformer (RDT) - PyTorch, Python

- Implemented a Recurrent Depth Transformer from scratch in PyTorch that loops one shared block instead of stacking layers; matched a 6-layer GPT baseline's multi-hop accuracy with 30-47% fewer parameters (extra loops did not extend accuracy past trained depths).
- Built a tool-calling agent on the fine-tuned model; caught that its eval reported 100% task completion without checking correctness, verified by hand that only 1 of 4 answers was right, and relabeled the metric to say what it actually measured.

GlassBox - OpenTelemetry, FastAPI, React, MCP

- Built an agent observability tool that records each model generation and tool call as OpenTelemetry spans and displays them in a FastAPI + React waterfall viewer.
- Used it to trace a multi-step failure in the RDT project's agent to repetition collapse on its second tool call (not a parsing bug), and prototyped a failure classifier that reuses confidence-referee's grounding signal unchanged.

SKILLS

ML: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, NumPy, Pandas

AI/LLM: Ollama, LLM Evaluation, Confidence Calibration, DistilBERT Fine-Tuning, Prompt Engineering, MCP

Databases: PostgreSQL, MySQL

Frontend: React, React Native, Aurelia, D3.js, HTML5, CSS3

Languages: Python, TypeScript, JavaScript, C#, SQL

Backend / APIs: ASP.NET, Node.js, FastAPI, RESTful APIs

Cloud: AWS, GCP

DevOps / Tools: Docker, Kubernetes, OpenTelemetry, Git/GitHub, Azure DevOps, Jira

PUBLICATION

Devkar, R. Prathibha, Daniel, M.S., John, N.P., et al. (2021). **Speaking Mouth System for Speech-Impaired People using Hand Gestures**

Built a custom 4,000-image dataset and a 2-layer CNN (96% test accuracy) for an end-to-end gesture-to-speech pipeline; published in the Springer FSST 2019 proceedings.

EDUCATION

Master of Science in Computer Science - The University of Texas at Arlington, USA

May 2026